Diode models

Practice questions · Day 03 · Mon 26 Oct · pack E2

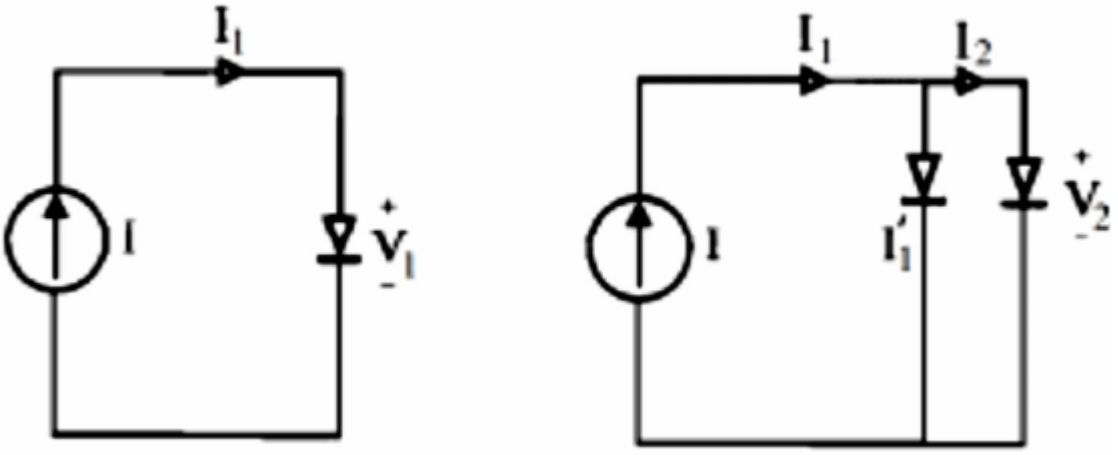
5. Practice

From your course files

WS2 Q8 (course question). A diode has a reverse saturation current of 0.2 pA, $n = 1$ and operates at room temperature. Find the voltage drop at the below currents:
i. 1 mA
ii. 10 mA
iii. 100 mA
iv. Explain how much the diode voltage drop changes for every order of magnitude increase in current. Discuss it in your group.

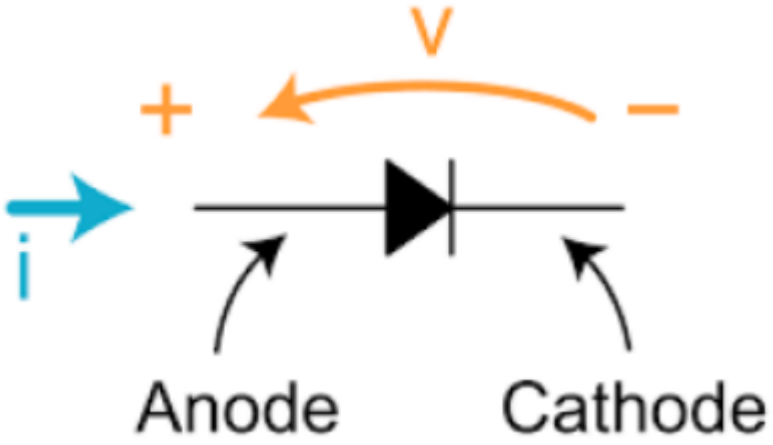
WS2 Q9 (course question). For a diode in the forward bias region at room temperature, we know that the current is 1 mA and 10 mA at 0.7 V and 0.8 V, respectively. What are the value of n and I_s for this diode?

WS2 Q10 (course question). The diode in the circuit below is energised by a current source I. If we add another similar diode parallel to the first one, how does the voltage change across the parallel diodes? Discuss it in your group.



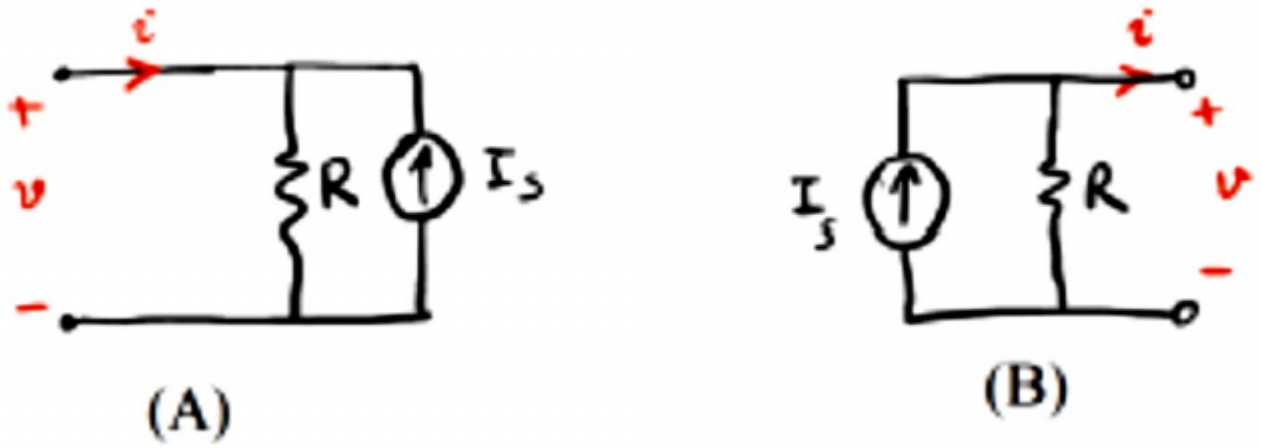
WS2 Fig. 2 (from the workshop sheet)

WS2 Q11 (course question). Consider a forward-biased diode. The end of the diode, which has the higher voltage, corresponds to [A] and its type is [B]. It is doped by [C]. The majority carriers are [D] and minority carriers are [E].



WS2 Fig. 3: diode symbol (from the workshop sheet)

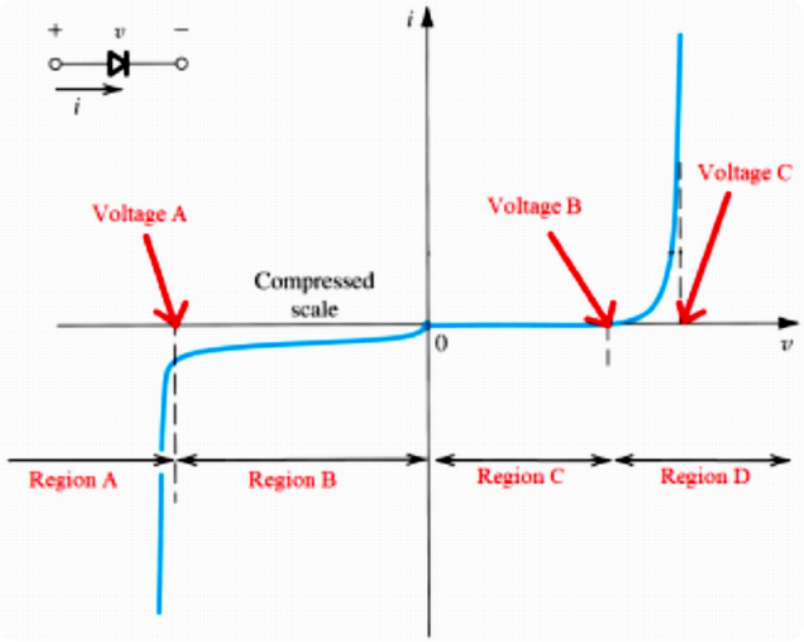
WS2 Q12 (course question). Do you expect a linear relationship between voltage and current in the two circuits below? Why? Discuss it in your group.



WS2 Fig. 4: circuits (A) and (B) (from the workshop sheet)

WS2 Q13 (course question). Now, plot the current (i)-voltage (v) characteristics curves for each of the two circuits in Fig. 4 based on the current direction and voltage polarity on the circuit and confirm your answer in Question 12.

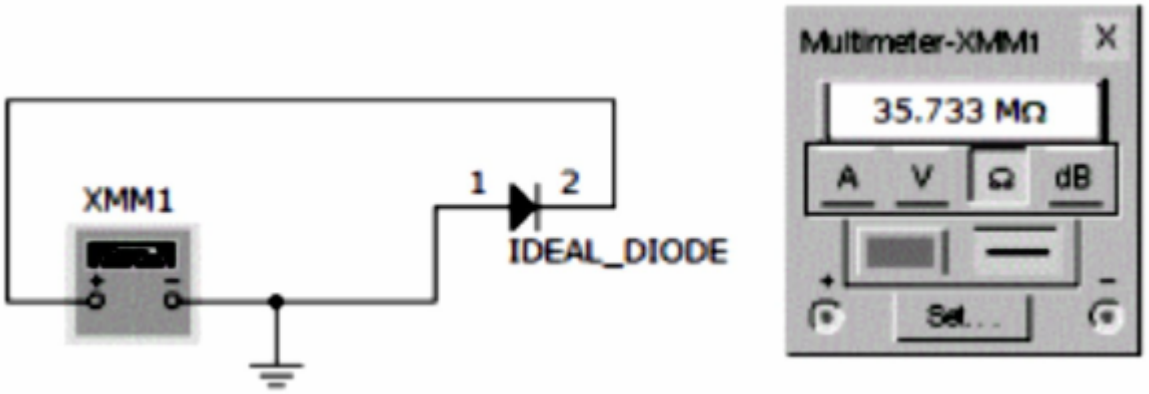
WS2 Q14 (course question). In the Fig. 5, full diode operational regions are shown according to the current direction and voltage polarities in the figure. Identify the regions and voltages mentioned in the figure. Discuss the diode's operation in each region in your group.



WS2 Fig. 5: diode operating regions (from the workshop sheet)

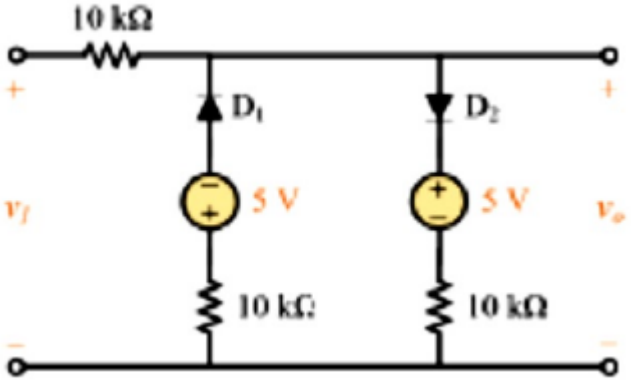
WS2 Q15 (course question). Considering the diode's operational regions in Question 14, we can linearize the diode's operation using the piece-wise linearization technique. How many pieces do you think we need? Discuss it in your group. *Note: Piecewise linear curve is a connected sequence of line segments. Piecewise linear function is a function whose domain can be decomposed into pieces on which the function is linear.*

WS2 Q16 (course question). In the circuit shown below, an ideal diode is connected to a multimeter. The device shows 35.733 MΩ. How do you explain this value? Is there anything wrong with the diode or our circuit? What do you expect to see if we rotate the diode connection? Discuss it in your group.



WS2 Fig. 6 (from the workshop sheet)

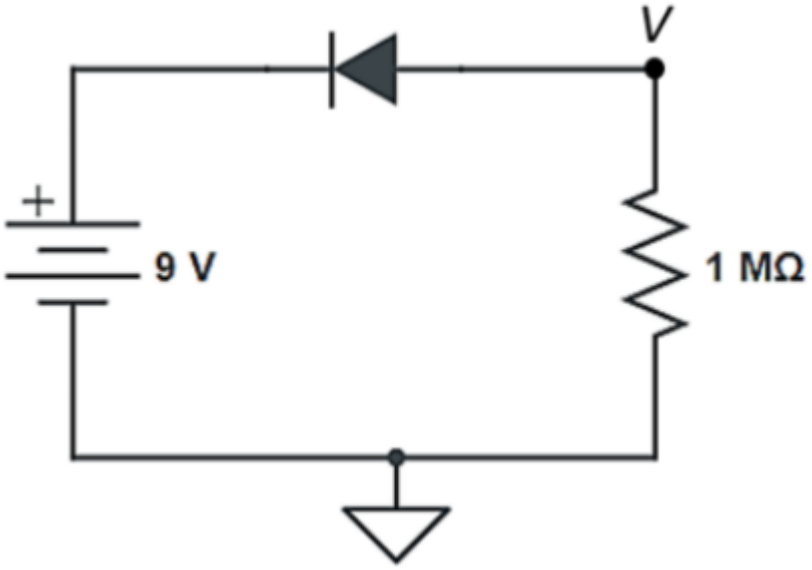
WS2 Q17 (course question). In the circuit below, replace the two diodes with their equivalent circuit model mentioned below in forward biasing (conducting) mode:
1. Ideal model
2. Voltage drop model, V_D
3. Voltage drop with resistor model, $V_D + R$



WS2 Fig. 8 (from the workshop sheet)

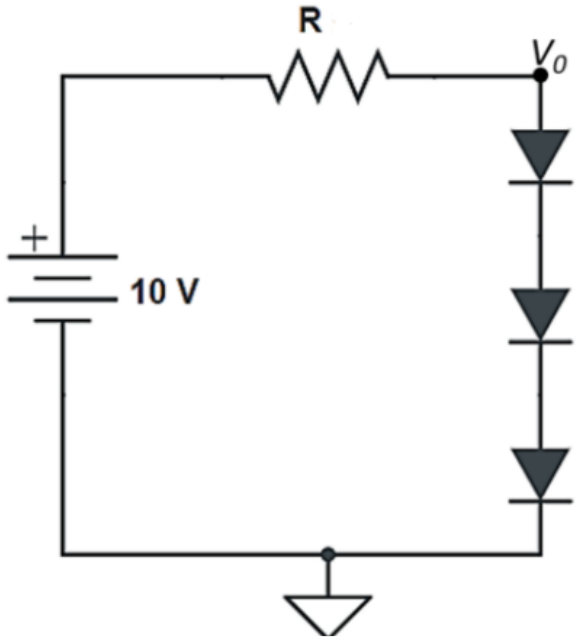
WS3 Q1 (course question). A silicon diode at 25 °C has a reverse saturation current of 1 µA. At what applied voltage will the diode current be 100 mA? Assume $n = 1$.

WS3 Q2 (course question). The diode circuit shown in Fig. 1 is a large, high-current device whose reverse current is reasonably independent of voltage, and doubles for every 10 °C rise in temperature. If $V = 1$ V at 20 °C, find the value of V at 40 °C and 0 °C.



WS3 Fig. 1 (from the workshop sheet)

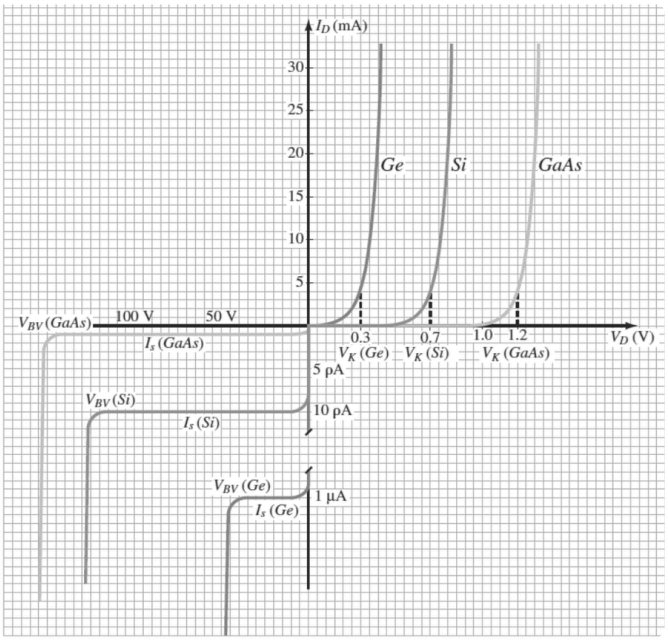
WS3 Q3 (course question). Design the circuit shown in Fig. 2 to provide an output voltage of 2.4 V. Assume that the diodes available have 0.7 V drop at 1 mA, and that such voltage varies by 0.1 V for every decade change in current.



WS3 Fig. 2 (from the workshop sheet)

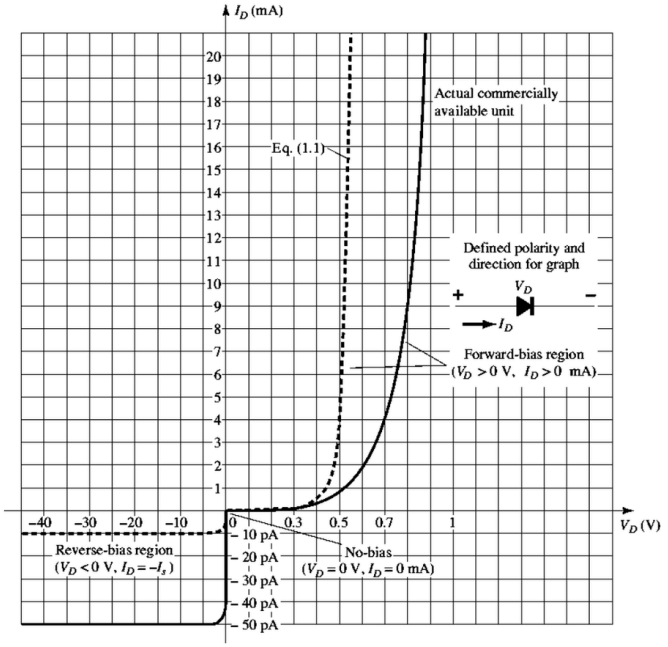
WS3 Q4 (course question). Determine the thermal voltage for a diode at a temperature of 20 °C. For the same diode, find the diode current if $I_s = 40$ nA, $n = 2$, and the applied bias voltage is 0.5 V.

WS3 Q5 (course question). Find the piecewise-linear equivalent circuit for the germanium and gallium arsenide diodes of Fig. 3.



WS3 Fig. 3 (from the workshop sheet)

WS3 Q6 (course question). The characteristics of a commercially available diode are shown in Fig. 4 below. Determine: (a) the static or dc resistance of the diode at a forward current of 4 mA. (b) the dynamic (ac) resistance at a forward current of 10 mA.



WS3 Fig. 4 (from the workshop sheet)

WS3 Q7 (course question). Determine the level of V_o for each network of Fig. 5 below:

WS3 Fig. 5 (a) and (b) (from the workshop sheet)

WS3 Q8 (course question). In the circuit shown in Fig. 6, find the voltages V_A and V_B using ideal diode model. (Assume $V_{threshold} = 0.7$ V)

WS3 Fig. 6 (from the workshop sheet)

PT Q2 (practice test, 5 marks). Consider a silicon diode with its exponential model and $n = 1.5$. Find the change in the voltage across it when the diode operates at 27 °C and the current changes from 0.1 mA to 10 mA.

PT Q3 (practice test, 8 marks). Consider the circuit below, which consists of the series combination of three identical diodes. Using the third piecewise linear model for the diodes with $V_D = 0.6$ V and $r_D = 0.20$ kΩ, answer the following questions. Assume that all diodes are **conducting**.
(a) Redraw the entire circuit using the third piecewise linear equivalent circuit model of diodes. You do not need to simplify the circuit. [2 marks]
(b) Verify whether all three diodes are **conducting** under the given circuit conditions. [4 marks]
(c) Find the output voltage, V_o, as shown on the circuit. [2 marks]

PT Q3 figure (from the practice-test paper)

New questions

N1 [4 marks] A diode with $n = 1.8$ operates at 50 °C and conducts 2 mA at 0.62 V.

a. Find V_T. [1]
b. Find V_D when the current falls to 0.2 mA. [2]
c. Find the small-signal resistance r_d at 2 mA. [1]

N2 [6 marks] At 27 °C a diode carries 0.5 mA at 0.60 V and 5 mA at 0.69 V.

a. Find n and I_S. [3]
b. Using your answer to (a), predict V_D at 20 mA. [1]
c. At 5 mA, find the dc resistance R_D and the small-signal resistance r_d, and say which one a 2 mV ac signal would see. [2]

N3 [4 marks] A silicon diode drops 0.65 V at 1 mA and has a reverse current of 10 nA, both at 25 °C. Use the course temperature rules.

a. Estimate V_D at 1 mA when the diode is at 75 °C. [1]
b. Find the reverse current at 75 °C and at 0 °C. [2]
c. The small-signal resistance at 1 mA ($n = 1$) changes between 25 °C and 75 °C. Give both values. [1]

N4 [7 marks] (multi-diode, iterate) A +10 V supply feeds a 1 kΩ resistor to node A. From A, diode D1 (anode at A) goes to node B, and B has a 2 kΩ resistor to ground. Also from A, diode D2 (anode at A) goes through a 1 kΩ resistor to the + terminal of an 8 V source whose − terminal is grounded. Use the constant-drop model, $V_D = 0.7$ V. V_o is the voltage at B.

a. Assume both diodes are ON. Solve, and show that the assumption fails. [3]
b. Correct the assumption, verify it, and find V_o. [4]

Circuit for N4.

N5 [8 marks] (two diodes, two input polarities, chained) V_i (+ at the top) feeds $R = 1$ kΩ to node A. V_o is the voltage at A. Three branches run from A to ground:

- **Branch 1:** D1 with its anode at A, then a 2 V battery with + facing D1, then 0.5 kΩ to ground.
- **Branch 2:** D2 with its **cathode** at A, its anode connected through 1 kΩ to ground.
- **Branch 3:** a 4 kΩ resistor.

Both diodes use the 3rd model with $V_D = 0.7$ V and $r_D = 0.1$ kΩ.

a. $V_i = +6$ V: state your assumptions, find V_o and the diode current, and verify. [3]
b. $V_i = -6$ V: repeat. [3]
c. Using the OFF-OFF circuit, find the range of V_i over which both diodes are OFF. [2]

Circuit for N5: three branches from A to ground.

N6 [8 marks] (OR gate with unequal inputs: the near-miss) $V_1 = 5$ V feeds 1 kΩ, then D1 (anode on the input side), to node V_o. V_2 feeds 1 kΩ, then D2, to the same node. V_o has 4 kΩ to ground. Both diodes use the 3rd model with $V_D = 0.7$ V and $r_D = 0.1$ kΩ.

a. Take $V_2 = 4$ V. Assume both diodes are ON, solve, and test the assumption. [3]
b. Correct the assumption, verify, and find V_o. [3]
c. Find the smallest V_2 at which D2 begins to conduct. [1]
d. Find V_o when $V_2 = 5$ V. [1]

Circuit for N6: diode OR gate with unequal inputs.

6. Quick recall check

1. With $n = 1$ and $V_T = 25$ mV, how much does V_D rise per decade of current? And with $n = 2$?
2. Two identical diodes are in parallel and driven by a current source. Does the voltage rise or fall compared with one diode, and by roughly how much?
3. What does each of the three piecewise-linear models become when the diode is OFF?
4. Find r_d at $I_D = 13$ mA for $n = 1$ and $V_T = 26$ mV.
5. A Si diode leaks 5 nA at 30 °C. What does it leak at 50 °C, and which way does its forward drop move?